3. Dictyostele:

- Stele is separated into several strands and each one is called meristele.
- It is found in *Adiantum*.

4. Eustele:

- Stele is split into distinct collateral vascular bundle around the pith.
- It is found in dicot stem.

Exercise : 18

Types of venation.

Aim: To study and identify the types of venation

Principle: The arrangement of veins and veinlets on the leaf blade or lamina is called venation. Internally, the vein contains vascular tissues. Conventionally venation is classified into two types namely, Reticulate venation and Parallel venation.

Requirements: Fresh specimen / photographs / pictures of pinnately reticulate, palmately reticulate convergent, palmetely reticulate divergent, pinnately parallel, palmatley parallel convergent and palmatley parallel divergent leaves.

Diagnostic features:

1. Pinnately reticulate:

- One midrib in the centre which forms many lateral branches to form a network.
- Example: *Mangifera, Nerium, Ficus*

2. Palmately reticulate convergent:

- Two or more main veins arising from a single point, all veins converge at the apex of leaf.
- Example: *Zizyphus, Cinnamomum*.

3. Palmately reticulate divergent:

- All principal veins originate from the base and diverge from one another towards the margin of the leaf.
- Example: *Cucurbita, Carica*.

4. Pinnately parallel:

- Prominent midrib in the centre, from which arise many veins perpendicularly and run parallel to each other.
- Example: *Musa*, Zinger, Canna.

5. Palmately Parallel Convergent:

- All principal run parallal to each other from the base of the lamina and join at the apex
- Example: Bamboo, Rice

6. Palmately Parallel Divergent:

- All principal veins originate from the base and diverge towards the margin as in Palm.
- Example: *Borassus flabellifer*.

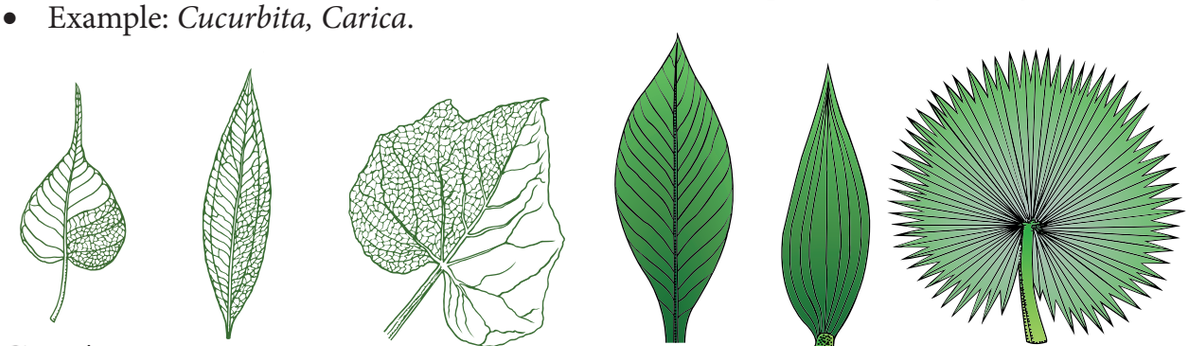


Figure 18: Types of Venation

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